HARD · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

Geometry and Trigonometry

11

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 D

D. The measures of angles M and T are $70°$ and $50°$, respectively.

Choice D is correct. Two triangles are similar if they have three pairs of congruent corresponding angles. It's given that angles L and R each measure $60°$, and so these corresponding angles are congruent. If angle M is $70°$, then angle N must be $50°$ so that the sum of the angles in triangle LMN is $180°$. If angle T is $50°$, then angle S must be $70°$ so that the sum of the angles in triangle RST is $180°$. Therefore, if the measures of angles M and T are $70°$ and $50°$, respectively, then corresponding angles M and S are both $70°$, and corresponding angles N and T are both $50°$. It follows that triangles LMN and RST have three pairs of congruent corresponding angles, and so the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle LMN is similar to triangle RST is that the measures of angles M and T are $70°$ and $50°$, respectively.

Choice A is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the two sides given are not proportional and the angle given is not included by the given sides.

Choice B is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the angle given is not included between the proportional sides.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2 D

D. 85

Choice D is correct. The volume, V, of a right circular cone is given by the formula $V = \frac{1}{3}\pi r^2 h$, where πr^2 is the area of the circular base of the cone and h is the height. It's given that this right circular cone has a volume of $71{,}148\pi$ cubic centimeters and the area of its base is $5{,}929\pi$ square centimeters. Substituting $71{,}148\pi$ for V and $5{,}929\pi$ for πr^2 in the formula $V = \frac{1}{3}\pi r^2 h$ yields $71{,}148\pi = \frac{1}{3}5{,}929\pi h$. Dividing each side of this equation by $5{,}929\pi$ yields $12 = \frac{h}{3}$. Multiplying each side of this equation by 3 yields $36 = h$. Let s represent the slant height, in centimeters, of this cone. A right triangle is formed by the radius, r, height, h, and slant height, s, of this cone, where r and h are the legs of the triangle and s is the hypotenuse. Using the Pythagorean theorem, the equation $r^2 + h^2 = s^2$ represents this relationship. Because $5{,}929\pi$ is the area of the base and the area of the base is πr^2, it follows that $5{,}929\pi = \pi r^2$. Dividing both sides of this equation by π yields $5{,}929 = r^2$. Substituting $5{,}929$ for r^2 and 36 for h in the equation $r^2 + h^2 = s^2$ yields $5{,}929 + 36^2 = s^2$, which is equivalent to $5{,}929 + 1{,}296 = s^2$, or $7{,}225 = s^2$. Taking the positive square root of both sides of this equation yields $85 = s$. Therefore, the slant height of the cone is 85 centimeters.

Choice A is incorrect. This is one-third of the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice B is incorrect. This is the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice C is incorrect. This is the radius, in centimeters, of the base, not the slant height, in centimeters, of this cone.

Q3 A

A. -3

Choice A is correct. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It's given that the circle has its center at $-4, -6$ and line k is tangent to the circle at the point $-7, -7$. The slope of a radius defined by the points q, r and s, t can be calculated as $\frac{t-r}{s-q}$. The points $-7, -7$ and $-4, -6$ define the radius of the circle at the point of tangency. Therefore, the slope of this radius can be calculated as $\frac{-6 - -7}{-4 - -7}$, or $\frac{1}{3}$. If a line and a radius are perpendicular, the slope of the line must be the negative reciprocal of the slope of the radius. The negative reciprocal of $\frac{1}{3}$ is -3. Thus, the slope of line k is -3.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the slope of the radius of the circle at the point of tangency, not the slope of line k.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

B

B. $\frac{5}{12}$

Choice B is correct. It's given that right triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. It's given that the side lengths of the right triangle RST are $RS = 20$, $ST = 48$, and $TR = 52$. Corresponding angles in similar triangles are equal. It follows that the measure of angle T is equal to the measure of angle W. The hypotenuse of a right triangle is the longest side. It follows that the hypotenuse of triangle RST is side TR. The hypotenuse of a right triangle is the side opposite the right angle. Therefore, angle S is a right angle. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle T is side ST. The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle T is side RS. The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore, $\tan T = \frac{RS}{ST}$. Substituting 20 for RS and 48 for ST in this equation yields $\tan T = \frac{20}{48}$, or $\tan T = \frac{5}{12}$. The tangents of two acute angles with equal measures are equal. Since the measure of angle T is equal to the measure of angle W, it follows that $\tan T = \tan W$. Substituting $\frac{5}{12}$ for $\tan T$ in this equation yields $\frac{5}{12} = \tan W$. Therefore, the value of $\tan W$ is $\frac{5}{12}$.

Choice A is incorrect. This is the value of $\sin W$.

Choice C is incorrect. This is the value of $\cos W$.

Choice D is incorrect. This is the value of $\frac{1}{\tan W}$.

Q5

4

The correct answer is 4.5. According to the properties of right triangles, BD divides triangle ABC into two similar triangles, ABD and BCD. The corresponding sides of ABD and BCD are proportional, so the ratio of BD to AD is the same as the ratio of DC to BD. Expressing this information as a proportion gives $\frac{6}{8} = \frac{DC}{6}$. Solving the proportion for DC results in $DC = 4.5$. Note that 4.5 and 9/2 are examples of ways to enter a correct answer.

Q6

C

C. 44

Choice C is correct. The volume of right circular cylinder A is given by the expression $\pi r^2 h$, where r is the radius of its circular base and h is its height. The volume of a cylinder with twice the radius and half the height of cylinder A is given by $\pi(2r)^2\left(\frac{1}{2}\right)h$, which is equivalent to $4\pi r^2\left(\frac{1}{2}\right)h = 2\pi r^2 h$. Therefore, the volume is twice the volume of cylinder A, or $2 \times 22 = 44$.

Choice A is incorrect and likely results from not multiplying the radius of cylinder A by 2. Choice B is incorrect and likely results from not squaring the 2 in 2r when applying the volume formula. Choice D is incorrect and likely results from a conceptual error.

Q7

A

A. $\sqrt{206}$

Choice A is correct. It's given that points A and B lie on the circle with center C. Therefore, $\overline{AC}$ and $\overline{BC}$ are both radii of the circle. Since all radii of a circle are congruent, $\overline{AC}$ is congruent to $\overline{BC}$. The length of $\overline{AC}$, or the distance from point A to point C, can be found using the distance formula, which gives the distance between two points, x_1, y_1 and x_2, y_2, as $\sqrt{x_1 - x_2{}^2 + y_1 - y_2{}^2}$. Substituting the given coordinates of point A, $h+1, k+\sqrt{102}$, for x_1, y_1 and the given coordinates of point C, h, k, for x_2, y_2 in the distance formula yields $\sqrt{h+1-h^2+k+\sqrt{102}-k^2}$, or $\sqrt{1^2+\sqrt{102}^{\,2}}$, which is equivalent to $\sqrt{1+102}$, or $\sqrt{103}$. Therefore, the length of $\overline{AC}$ is $\sqrt{103}$ and the length of $\overline{BC}$ is $\sqrt{103}$. It's given that angle ACB is a right angle. Therefore, triangle ACB is a right triangle with legs $\overline{AC}$ and $\overline{BC}$ and hypotenuse $\overline{AB}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2+b^2=c^2$. Substituting $\sqrt{103}$ for a and b in this equation yields $\sqrt{103}^{\,2}+\sqrt{103}^{\,2}=c^2$, or $103+103=c^2$, which is equivalent to $206=c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{206}=c$. Therefore, the length of $\overline{AB}$ is $\sqrt{206}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the length of $\overline{AB}$ if the length of $\overline{AC}$ were 103, not $\sqrt{103}$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

.6956

Accept also: `.6957`, `16/23`

The correct answer is $\frac{16}{23}$. In a right triangle, the sine of an acute angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the length of the side opposite the angle with measure $x°$ is 16 units and the length of the hypotenuse is 23 units. Therefore, the value of $\sin x°$ is $\frac{16}{23}$. Note that 16/23, .6956, .6957, 0.695, and 0.696 are examples of ways to enter a correct answer.

Q9

D

D. $V = \pi r^3 + 4\pi r^2$

Choice D is correct. The volume, V, of a right cylinder is given by the formula $V = \pi r^2 h$, where r represents the radius of the base of the cylinder and h represents the height. Since the height is 4 inches longer than the radius, the expression $r+4$ represents the height of each cylindrical container. It follows that the volume of each container is represented by the equation $V = \pi r^2(r+4)$. Distributing the expression πr^2 into each term in the parentheses yields $V = \pi r^3 + 4\pi r^2$.

Choice A is incorrect and may result from representing the height as $4r$ instead of $r+4$. Choice B is incorrect and may result from representing the height as $2r$ instead of $r+4$. Choice C is incorrect and may result from representing the volume of a right cylinder as $V = \pi r h$ instead of $V = \pi r^2 h$.

Q10

100

The correct answer is 100. An equation of a circle in the *xy*-plane can be written as $(x-t)^2 + (y-u)^2 = r^2$, where the center of the circle is (t, u), the radius of the circle is r, and where t, u, and r are constants. It's given that the equation of circle A is $(x+5)^2 + (y-5)^2 = 25$, which is equivalent to $(x+5)^2 + (y-5)^2 = 5^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is 5. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is $2(5)$, or 10. Substituting -5 for t, 5 for u, and 10 for r in the equation $(x-t)^2 + (y-u)^2 = r^2$ yields $(x+5)^2 + (y-5)^2 = 10^2$, which is equivalent to $(x+5)^2 + (y-5)^2 = 100$. It follows that the equation of circle B in the *xy*-plane is $(x+5)^2 + (y-5)^2 = 100$. It's also given that the equation defining circle B in the *xy*-plane is $(x+5)^2 + (y-5)^2 = k$, where k is a constant. Therefore, the value of k is 100.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank